03. Wireshark Capture Filters and Display Filters

1. What Is Wireshark?

Definition: Wireshark is a free, open-source network protocol analyzer used to capture and inspect network packets.

Key points:

- Helps troubleshoot network problems.
- Displays packet source, destination, protocols and timestamps.
- Helps investigate suspicious traffic.
- Supports capture filters and display filters.

2. Capture Filters

Definition: Capture filters select which packets Wireshark records while a capture is running.

Capture filters use Berkeley Packet Filter (BPF) syntax.

Examples:

- tcp — captures TCP traffic.
- udp — captures UDP traffic.
- port 53 — captures traffic using port 53.
- host 192.168.1.10 — captures traffic to or from that host.

3. Display Filters

Definition: Display filters select which packets are shown from the packets already captured.

They use Wireshark's own display-filter syntax.

Examples:

- dns — displays DNS packets.
- tcp — displays TCP packets.
- icmp — displays ICMP packets.
- ip.addr == 192.168.1.10 — displays IPv4 packets to or from that address.

4. Differences

Feature	Capture Filter	Display Filter
When applied	Before/during capture	After packets are captured
Purpose	Controls which packets are recorded	Controls which packets are displayed
Syntax	BPF	Wireshark display-filter syntax
Can exclude packets from the saved capture?	Yes	No, it only hides packets from the current view
Example	port 53	dns

5. Ten Working Examples

No.	Capture Filter	Display Filter
1	tcp	tcp
2	udp	udp
3	port 53	dns
4	port 80	tcp.port == 80
5	port 443	tcp.port == 443
6	icmp	icmp
7	arp	arp
8	host 192.168.1.10	ip.addr == 192.168.1.10
9	net 192.168.1.0/24	ip.addr == 192.168.1.10
10	tcp port 445	tcp.port == 445

Replace example IP addresses with addresses from your own lab. The display filter in row 9 shows one host in the subnet, whereas the capture filter captures traffic involving the whole subnet.

6. Conclusion

Capture filters control what gets recorded, while display filters help investigate packets that have already been captured. Both are important for network troubleshooting and SOC investigations.